Selenium WebDriver

100 Interview Questions, Answers & Java Coding Examples

Senior SDET / Test Automation Lead — Worldline Round 2

Expanded answers based on your uploaded interview guide. Every question includes a relevant implementation pattern.

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Note: Code blocks use a black background with light text for readability. Snippets are illustrative; add relevant imports and integrate them into your framework.

1. Selenium Fundamentals

Q1. What is Selenium?

Interview-ready answer: An open-source browser-automation ecosystem. WebDriver provides APIs to control real browsers. It is mainly for web UI automation. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); }

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q2. What are Selenium's main components?

Interview-ready answer: WebDriver, Selenium Grid and the Selenium IDE ecosystem. For senior roles, WebDriver and Grid are the most important. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); }

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q3. What is WebDriver?

Interview-ready answer: The API/protocol-based interface used by automation code to control a browser. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); }

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q4. Explain Selenium WebDriver architecture.

Interview-ready answer: Test code → Selenium WebDriver API → browser driver/WebDriver implementation → browser → web application. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); }

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q5. Why is modern WebDriver preferred over Selenium RC?

Interview-ready answer: It uses modern browser automation standards and provides a cleaner, more reliable architecture than the legacy RC approach. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); } Worldline Selenium Interview Guide | Java Page 3

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q6. What is the W3C WebDriver standard?

Interview-ready answer: A standard browser-automation protocol defining communication between automation clients and browser implementations. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); }

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q7. What is Selenium Manager?

Interview-ready answer: Selenium's built-in driver/browser management capability that can reduce manual driver setup in modern Selenium. Java / implementation example public final class DriverFactory { private DriverFactory() {} public static WebDriver create() { ChromeOptions options = new ChromeOptions(); options.addArguments("--start-maximized"); return new ChromeDriver(options); } }

Java / implementation example

```
public final class DriverFactory {
    private DriverFactory() {}

    public static WebDriver create() {
        ChromeOptions options = new ChromeOptions();
        options.addArguments("--start-maximized");
        return new ChromeDriver(options);
    }
}
```

Q8. Selenium vs Playwright?

Interview-ready answer: Selenium has a mature WebDriver ecosystem and Grid; Playwright provides modern browser control, auto-waiting and browser-context isolation. Choose based on requirements. Java / implementation example // Selenium Java: wait.until(ExpectedConditions.elementToBeClickable(By.id("save"))).click(); // Playwright Java equivalent: // page.locator("#save").click();

Java / implementation example

```
// Selenium Java:
wait.until(ExpectedConditions.elementToBeClickable(By.id("save"))).click();
// Playwright Java equivalent:
// page.locator("#save").click();
```

Q9. Can Selenium automate native mobile apps?

Interview-ready answer: No. Selenium targets web browsers; Appium is commonly used for native/hybrid mobile applications. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); }

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

Q10. What are Selenium limitations?

Interview-ready answer: Web-focused, requires good synchronisation/locator/data design, and does not itself provide a complete test runner or reporting platform. Java / implementation example WebDriver driver = new ChromeDriver(); try { driver.get("https://example.com"); System.out.println(driver.getTitle()); } finally { driver.quit(); } Worldline Selenium Interview Guide | Java Page 4

Java / implementation example

```
WebDriver driver = new ChromeDriver();
try {
    driver.get("https://example.com");
    System.out.println(driver.getTitle());
} finally {
    driver.quit();
}
```

2. Locators & WebElements

Q11. What locator strategies are available?

Interview-ready answer: id, name, class name, tag name, link text, partial link text, CSS and XPath. Prefer stable test attributes where available. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q12. CSS vs XPath?

Interview-ready answer: CSS is concise and readable; XPath is more expressive for relationships/text. Choose the simplest stable locator. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q13. Why avoid absolute XPath?

Interview-ready answer: It depends on DOM hierarchy and breaks easily when markup changes. Worldline Round 2 — Selenium Comprehensive Interview Preparation Page 4 Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q14. What makes a good locator?

Interview-ready answer: Stable, unique, readable and based on meaningful attributes; data-testid is often excellent. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q15. findElement vs findElements?

Interview-ready answer: findElement returns one element and throws if absent; findElements returns a list and returns an empty list when nothing matches. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q16. How do you handle dynamic elements?

Interview-ready answer: Use stable attributes/relative conditions and explicit waits for the required state. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q17. isDisplayed vs isEnabled vs isSelected?

Interview-ready answer: They validate visibility, enabled state and selection state respectively. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click();

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

Q18. What is WebElement?

Interview-ready answer: A Selenium representation of a DOM element with actions and state methods. Java / implementation example By submit = By.cssSelector("[data-testid='submit']"); WebElement button = driver.findElement(submit); button.click(); Worldline Selenium Interview Guide | Java Page 5

Java / implementation example

```
By submit = By.cssSelector("[data-testid='submit']");
WebElement button = driver.findElement(submit);
button.click();
```

3. Waits & Synchronisation

Q19. Why do we need waits?

Interview-ready answer: Modern UIs are asynchronous; elements can appear, become enabled or be replaced later. Wait for meaningful conditions. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q20. Implicit vs explicit wait?

Interview-ready answer: Implicit is global for element lookup; explicit targets a specific condition. Prefer a deliberate explicit-wait strategy. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q21. Why avoid Thread.sleep()?

Interview-ready answer: Fixed delays waste time when the app is ready early and can still be too short when it is slow. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q22. What is FluentWait?

Interview-ready answer: A configurable wait with timeout, polling interval and ignored exceptions. Java / implementation example public final class WaitUtils { private final WebDriverWait wait; public WaitUtils(WebDriver driver) { wait = new WebDriverWait(driver, Duration.ofSeconds(10)); } public WebElement visible(By locator) { return wait.until(ExpectedConditions.visibilityOfElementLocated(locator)); } }

Java / implementation example

```
public final class WaitUtils {
    private final WebDriverWait wait;

    public WaitUtils(WebDriver driver) {
        wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    }

    public WebElement visible(By locator) {
        return wait.until(
            ExpectedConditions.visibilityOfElementLocated(locator));
    }
}
```

Q23. What is ExpectedConditions?

Interview-ready answer: Reusable conditions for visibility, presence, clickability, alerts, frames, titles and more. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q24. Presence vs visibility?

Interview-ready answer: Presence means in DOM; visibility means present and visible. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q25. What does elementToBeClickable mean?

Interview-ready answer: Visible and enabled; it does not guarantee that an overlay cannot intercept the click. Java / implementation example wait.until(ExpectedConditions.invisibilityOfElementLocated(By.cssSelector(".loading-overlay"))); wait.until(ExpectedConditions.elementToBeClickable(By.id("confirm"))).click(); Worldline Selenium Interview Guide | Java Page 6

Java / implementation example

```
wait.until(ExpectedConditions.invisibilityOfElementLocated(
    By.cssSelector(".loading-overlay")));
wait.until(ExpectedConditions.elementToBeClickable(By.id("confirm"))).click();
```

Q26. How do you build a reusable wait utility?

Interview-ready answer: Centralise WebDriverWait methods so page objects do not duplicate synchronisation logic. Java / implementation example public final class WaitUtils { private final WebDriverWait wait; public WaitUtils(WebDriver driver) { wait = new WebDriverWait(driver, Duration.ofSeconds(10)); } public WebElement visible(By locator) { return wait.until(ExpectedConditions.visibilityOfElementLocated(locator)); } }

Java / implementation example

```
public final class WaitUtils {
    private final WebDriverWait wait;

    public WaitUtils(WebDriver driver) {
        wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    }

    public WebElement visible(By locator) {
        return wait.until(
            ExpectedConditions.visibilityOfElementLocated(locator));
    }
}
```

Q27. Should implicit and explicit waits be mixed?

Interview-ready answer: Prefer not to; mixing can make timing harder to reason about. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q28. How do you handle AJAX-driven UI changes?

Interview-ready answer: Wait for a state change such as element visibility, spinner disappearance, text/state change or URL change. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click(); Worldline Selenium Interview Guide | Java Page 7

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

4. Exceptions & Debugging

Q29. What is NoSuchElementException? Worldline Round 2 — Selenium Comprehensive Interview Preparation Page 5

Interview-ready answer: The locator did not find an element in the current page/context. Check locator, timing, iframe, page state and DOM. Java / implementation example try { wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("receipt"))); } catch (TimeoutException e) { throw new AssertionError("Receipt did not appear after checkout", e); }

Java / implementation example

```
try {
    wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("receipt")));
} catch (TimeoutException e) {
    throw new AssertionError("Receipt did not appear after checkout", e);
}
```

Q30. What is StaleElementReferenceException?

Interview-ready answer: The DOM changed and the stored element reference is no longer valid. Re-locate and synchronise. Java / implementation example By status = By.id("status"); // Re-locate after the DOM updates; do not reuse an old WebElement. WebElement current = wait.until(ExpectedConditions.visibilityOfElementLocated(status)); System.out.println(current.getText());

Java / implementation example

```
By status = By.id("status");
// Re-locate after the DOM updates; do not reuse an old WebElement.
WebElement current = wait.until(
    ExpectedConditions.visibilityOfElementLocated(status));
System.out.println(current.getText());
```

Q31. How do you handle ElementClickInterceptedException?

Interview-ready answer: Find the blocking overlay/state, wait for it to disappear and then interact normally. Java / implementation example wait.until(ExpectedConditions.invisibilityOfElementLocated(By.cssSelector(".loading-overlay"))); wait.until(ExpectedConditions.elementToBeClickable(By.id("confirm"))).click();

Java / implementation example

```
wait.until(ExpectedConditions.invisibilityOfElementLocated(
    By.cssSelector(".loading-overlay")));
wait.until(ExpectedConditions.elementToBeClickable(By.id("confirm"))).click();
```

Q32. What causes ElementNotInteractableException?

Interview-ready answer: The element exists but is hidden, disabled or otherwise not actionable. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q33. How do you debug a flaky Selenium test?

Interview-ready answer: Capture screenshot/source/logs, check timing, locators, data, environment, parallelism and application behaviour, then classify root cause. Java / implementation example try { runTestFlow(); } catch (RuntimeException e) { byte[] png = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES); Files.write(Path.of("target/failure.png"), png); Files.writeString(Path.of("target/page-source.html"), driver.getPageSource()); throw e; }

Java / implementation example

```
try {
    runTestFlow();
} catch (RuntimeException e) {
    byte[] png = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES);
    Files.write(Path.of("target/failure.png"), png);
    Files.writeString(Path.of("target/page-source.html"), driver.getPageSource());
    throw e;
}
```

Q34. Should we catch all Selenium exceptions?

Interview-ready answer: No. Catch only when meaningful recovery/context exists; otherwise fail clearly and preserve evidence. Java / implementation example try { wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("receipt"))); } catch (TimeoutException e) { throw new AssertionError("Receipt did not appear after checkout", e); }

Java / implementation example

```
try {
    wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("receipt")));
} catch (TimeoutException e) {
    throw new AssertionError("Receipt did not appear after checkout", e);
}
```

Q35. How do you handle TimeoutException?

Interview-ready answer: Treat it as evidence that the condition was not reached. Investigate condition, locator, app, environment and timing before increasing timeout. Java / implementation example System.out.println("URL: " + driver.getCurrentUrl()); System.out.println("Title: " + driver.getTitle()); Files.writeString(Path.of("target/page.html"), driver.getPageSource()); Worldline Selenium Interview Guide | Java Page 8

Java / implementation example

```
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + driver.getTitle());
Files.writeString(Path.of("target/page.html"), driver.getPageSource());
```

Q36. How do you avoid hiding failures in utilities?

Interview-ready answer: Do not swallow exceptions or return false success. Log useful context and rethrow/fail when recovery is impossible. Java / implementation example try { wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("receipt"))); } catch (TimeoutException e) { throw new AssertionError("Receipt did not appear after checkout", e); } Worldline Selenium Interview Guide | Java Page 9

Java / implementation example

```
try {
    wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("receipt")));
} catch (TimeoutException e) {
    throw new AssertionError("Receipt did not appear after checkout", e);
}
```

5. Windows, Frames, Alerts & Browser Controls

Q37. How do you handle multiple windows/tabs?

Interview-ready answer: Capture handles, wait for the new handle, switch deliberately, then return to the required handle. Java / implementation example String parent = driver.getWindowHandle(); driver.findElement(By.id("open-window")).click(); wait.until(d -> d.getWindowHandles().size() > 1); for (String handle : driver.getWindowHandles()) { if (!handle.equals(parent)) { driver.switchTo().window(handle); break; } }

Java / implementation example

```
String parent = driver.getWindowHandle();
driver.findElement(By.id("open-window")).click();
wait.until(d -> d.getWindowHandles().size() > 1);
for (String handle : driver.getWindowHandles()) {
    if (!handle.equals(parent)) {
        driver.switchTo().window(handle);
        break;
    }
}
```

Q38. How do you switch to an iframe?

Interview-ready answer: driver.switchTo().frame(...), interact, then defaultContent() or parentFrame(). Java / implementation example wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt(By.id("payment-frame"))); driver.findElement(By.id("card-number")).sendKeys("4111111111111111"); driver.switchTo().defaultContent();

Java / implementation example

```
wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt(
    By.id("payment-frame")));
driver.findElement(By.id("card-number")).sendKeys("4111111111111111");
driver.switchTo().defaultContent();
```

Q39. How do you handle nested iframes?

Interview-ready answer: Switch parent frame first, then child frame; unwind the context deliberately. Java / implementation example wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt(By.id("payment-frame"))); driver.findElement(By.id("card-number")).sendKeys("4111111111111111"); driver.switchTo().defaultContent();

Java / implementation example

```
wait.until(ExpectedConditions.frameToBeAvailableAndSwitchToIt(
    By.id("payment-frame")));
driver.findElement(By.id("card-number")).sendKeys("4111111111111111");
driver.switchTo().defaultContent();
```

Q40. How do you handle JavaScript alerts?

Interview-ready answer: Wait for alert presence, switch to Alert, read text and accept/dismiss/sendKeys as required. Java / implementation example Alert alert = wait.until(ExpectedConditions.alertIsPresent()); String text = alert.getText(); alert.accept(); System.out.println(text);

Java / implementation example

```
Alert alert = wait.until(ExpectedConditions.alertIsPresent());
String text = alert.getText();
alert.accept();
System.out.println(text);
```

Q41. How do you handle navigation?

Interview-ready answer: Use get/navigate and synchronise with the resulting application state. Java / implementation example driver.navigate().to("https://example.com/dashboard"); wait.until(ExpectedConditions.urlContains("/dashboard")); wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("dashboard")));

Java / implementation example

```
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + driver.getTitle());
Files.writeString(Path.of("target/page.html"), driver.getPageSource());
```

Q42. How do you handle authentication pop-ups?

Interview-ready answer: Use an application/browser-supported authentication strategy; avoid insecure credentials embedded in URLs. Java / implementation example String username = System.getenv("TEST_USERNAME"); String password = System.getenv("TEST_PASSWORD"); if (username == null || password == null) { throw new IllegalStateException("Configure test credentials in CI secrets"); }

Java / implementation example

```
String username = System.getenv("TEST_USERNAME");
String password = System.getenv("TEST_PASSWORD");
if (username == null || password == null) {
    throw new IllegalStateException("Configure test credentials in CI secrets");
}
```

Q43. How do you handle dropdowns?

Interview-ready answer: Use Select for native HTML select elements; interact with custom dropdown DOM for JavaScript widgets. Java / implementation example Select country = new Select(driver.findElement(By.id("country"))); country.selectByVisibleText("India");

Java / implementation example

```
Select country = new Select(driver.findElement(By.id("country")));
country.selectByVisibleText("India");
```

Q44. How do you upload a file? Worldline Round 2 — Selenium Comprehensive Interview Preparation Page 6

Interview-ready answer: For input[type=file], send the absolute path to the file input. Java / implementation example Path file = Path.of("src/test/resources/sample.pdf").toAbsolutePath(); driver.findElement(By.cssSelector("input[type='file']")).sendKeys(file.toString()); Worldline Selenium Interview Guide | Java Page 10

Java / implementation example

```
Path file = Path.of("src/test/resources/sample.pdf").toAbsolutePath();
driver.findElement(By.cssSelector("input[type='file']")).sendKeys(file.toString());
```

6. Actions, JavaScript, Cookies, Screenshots & Shadow DOM

Q45. What is Actions used for?

Interview-ready answer: Advanced mouse/keyboard interactions such as hover, drag-and-drop, click-and-hold and key combinations. Java / implementation example Actions actions = new Actions(driver); actions.moveToElement(driver.findElement(By.id("menu"))).perform(); actions.keyDown(Keys.CONTROL).sendKeys("a").keyUp(Keys.CONTROL).perform();

Java / implementation example

```
Actions actions = new Actions(driver);
actions.moveToElement(driver.findElement(By.id("menu"))).perform();
actions.keyDown(Keys.CONTROL).sendKeys("a").keyUp(Keys.CONTROL).perform();
```

Q46. When would you use JavaScriptExecutor?

Interview-ready answer: For browser-side tasks such as scrolling or inspection when normal WebDriver APIs are insufficient. Prefer normal WebDriver clicks first. Java / implementation example WebElement target = driver.findElement(By.id("submit")); ((JavascriptExecutor) driver).executeScript("arguments[0].scrollIntoView({block:'center'});", target);

Java / implementation example

```
WebElement target = driver.findElement(By.id("submit"));
((JavascriptExecutor) driver).executeScript(
    "arguments[0].scrollIntoView({block:'center'});", target);
```

Q47. How do you scroll to an element?

Interview-ready answer: Use Selenium interaction/scrolling or JS scrollIntoView where appropriate. Java / implementation example WebElement target = driver.findElement(By.id("submit")); ((JavascriptExecutor) driver).executeScript("arguments[0].scrollIntoView({block:'center'});", target);

Java / implementation example

```
WebElement target = driver.findElement(By.id("submit"));
((JavascriptExecutor) driver).executeScript(
    "arguments[0].scrollIntoView({block:'center'});", target);
```

Q48. How do you work with cookies?

Interview-ready answer: getCookies, addCookie, getCookieNamed, deleteCookie and deleteAllCookies. Java / implementation example driver.get("https://example.com"); driver.manage().addCookie(new Cookie("theme", "dark")); Cookie theme = driver.manage().getCookieNamed("theme"); System.out.println(theme == null ? "missing" : theme.getValue());

Java / implementation example

```
driver.get("https://example.com");
driver.manage().addCookie(new Cookie("theme", "dark"));
Cookie theme = driver.manage().getCookieNamed("theme");
System.out.println(theme == null ? "missing" : theme.getValue());
```

Q49. How do you take screenshots?

Interview-ready answer: Use TakesScreenshot and integrate automatic capture with the test listener/hook. Java / implementation example byte[] image = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES); Files.write(Path.of("target/screenshot.png"), image);

Java / implementation example

```
byte[] image = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES);
Files.write(Path.of("target/screenshot.png"), image);
```

Q50. How do you capture screenshots on failure?

Interview-ready answer: Use TestNG/JUnit listener or equivalent hook and attach evidence to the report. Java / implementation example byte[] image = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES); Files.write(Path.of("target/screenshot.png"), image);

Java / implementation example

```
byte[] image = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES);
Files.write(Path.of("target/screenshot.png"), image);
```

Q51. How do you capture page source/browser evidence?

Interview-ready answer: getPageSource plus browser logging/CDP/BiDi mechanisms where supported. Java / implementation example System.out.println("URL: " + driver.getCurrentUrl()); System.out.println("Title: " + driver.getTitle()); Files.writeString(Path.of("target/page.html"), driver.getPageSource());

Java / implementation example

```
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + driver.getTitle());
Files.writeString(Path.of("target/page.html"), driver.getPageSource());
```

Q52. What is Shadow DOM?

Interview-ready answer: Encapsulated DOM content. Open shadow roots can be accessed through Selenium's ShadowRoot/SearchContext APIs. Java / implementation example WebElement host = driver.findElement(By.cssSelector("user-card")); SearchContext shadow = host.getShadowRoot(); System.out.println(shadow.findElement(By.cssSelector(".display-name")).getText()); Worldline Selenium Interview Guide | Java Page 11

Java / implementation example

```
WebElement host = driver.findElement(By.cssSelector("user-card"));
SearchContext shadow = host.getShadowRoot();
System.out.println(shadow.findElement(By.cssSelector(".display-name")).getText());
```

7. Selenium 4 & Modern WebDriver

Q53. Important Selenium 4 features?

Interview-ready answer: W3C alignment, relative locators, Grid 4, new tab/window APIs, improved DevTools integration and Selenium Manager. Java / implementation example String original = driver.getWindowHandle(); driver.switchTo().newWindow(WindowType.TAB); driver.get("https://example.com/help"); driver.switchTo().window(original);

Java / implementation example

```
String original = driver.getWindowHandle();
driver.switchTo().newWindow(WindowType.TAB);
driver.get("https://example.com/help");
driver.switchTo().window(original);
```

Q54. What are relative locators?

Interview-ready answer: Spatial relationships such as above, below, toLeftOf, toRightOf and near. Java / implementation example WebElement label = driver.findElement(By.id("email-label")); WebElement input = driver.findElement(RelativeLocator.with(By.tagName("input")).below(label));

Java / implementation example

```
WebElement label = driver.findElement(By.id("email-label"));
WebElement input = driver.findElement(
    RelativeLocator.with(By.tagName("input")).below(label));
```

Q55. How do you open a new tab/window in Selenium 4?

Interview-ready answer: Use driver.switchTo().newWindow(WindowType.TAB/WINDOW). Java / implementation example String original = driver.getWindowHandle(); driver.switchTo().newWindow(WindowType.TAB); driver.get("https://example.com/help"); driver.switchTo().window(original);

Java / implementation example

```
String original = driver.getWindowHandle();
driver.switchTo().newWindow(WindowType.TAB);
driver.get("https://example.com/help");
driver.switchTo().window(original);
```

Q56. What is Selenium Grid 4?

Interview-ready answer: A distributed Selenium infrastructure for running WebDriver sessions across machines/browsers. Java / implementation example URL gridUrl = URI.create("http://localhost:4444").toURL(); WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions()); driver.get("https://example.com");

Java / implementation example

```
URL gridUrl = URI.create("http://localhost:4444").toURL();
WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions());
driver.get("https://example.com");
```

Q57. Why use Grid?

Interview-ready answer: Parallel execution, cross-browser coverage and scalable execution. Java / implementation example URL gridUrl = URI.create("http://localhost:4444").toURL(); WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions()); driver.get("https://example.com");

Java / implementation example

```
URL gridUrl = URI.create("http://localhost:4444").toURL();
WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions());
driver.get("https://example.com");
```

Q58. What is CDP?

Interview-ready answer: Chrome DevTools Protocol; it enables browser-specific capabilities such as network or browser inspection. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q59. What is WebDriver BiDi? Worldline Round 2 — Selenium Comprehensive Interview Preparation Page 7

Interview-ready answer: A bidirectional browser automation protocol designed to provide richer event-driven capabilities alongside WebDriver. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed()); Worldline Selenium Interview Guide | Java Page 12

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q60. Grid vs cloud browser providers?

Interview-ready answer: Grid gives infrastructure control; cloud providers offer managed scalable browser infrastructure. Decide based on security, cost, coverage and operations. Java / implementation example URL gridUrl = URI.create("http://localhost:4444").toURL(); WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions()); driver.get("https://example.com"); Worldline Selenium Interview Guide | Java Page 13

Java / implementation example

```
URL gridUrl = URI.create("http://localhost:4444").toURL();
WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions());
driver.get("https://example.com");
```

8. POM & PageFactory

Q61. What is Page Object Model?

Interview-ready answer: A design pattern that encapsulates page locators and interactions so tests remain business-readable and maintainable. Java / implementation example public class LoginPage { private final WebDriver driver; private final By username = By.id("username"); private final By password = By.id("password"); private final By login = By.id("login"); public LoginPage(WebDriver driver) { this.driver = driver; } public void login(String user, String pass) { driver.findElement(username).sendKeys(user); driver.findElement(password).sendKeys(pass); driver.findElement(login).click(); } }

Java / implementation example

```
public class LoginPage {
    private final WebDriver driver;
    private final By username = By.id("username");
    private final By password = By.id("password");
    private final By login = By.id("login");

    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }

    public void login(String user, String pass) {
        driver.findElement(username).sendKeys(user);
        driver.findElement(password).sendKeys(pass);
        driver.findElement(login).click();
    }
}
```

Q62. What makes a good Page Object?

Interview-ready answer: Stable locators, business-level actions, minimal duplication and clear separation of UI interaction from test intent. Java / implementation example public class LoginPage { private final WebDriver driver; private final By username = By.id("username"); private final By password = By.id("password"); private final By login = By.id("login"); public LoginPage(WebDriver driver) { this.driver = driver; } public void login(String user, String pass) { driver.findElement(username).sendKeys(user); driver.findElement(password).sendKeys(pass); driver.findElement(login).click(); } }

Java / implementation example

```
public class HeaderComponent {
    private final WebDriver driver;
    private final By accountMenu = By.id("account-menu");

    public HeaderComponent(WebDriver driver) {
        this.driver = driver;
    }

    public void openAccountMenu() {
        driver.findElement(accountMenu).click();
    }
}
```

Q63. POM vs PageFactory?

Interview-ready answer: POM is a design pattern; PageFactory is a Selenium support mechanism for @FindBy element fields. Java / implementation example public class LoginPage { private final WebDriver driver; private final By username = By.id("username"); private final By password = By.id("password"); private final By login = By.id("login"); public LoginPage(WebDriver driver) { this.driver = driver; } public void login(String user, String pass) { driver.findElement(username).sendKeys(user); driver.findElement(password).sendKeys(pass); driver.findElement(login).click(); } }

Java / implementation example

```
public class LoginPage {
    private final WebDriver driver;
    private final By username = By.id("username");
    private final By password = By.id("password");
    private final By login = By.id("login");

    public LoginPage(WebDriver driver) {
        this.driver = driver;
    }

    public void login(String user, String pass) {
        driver.findElement(username).sendKeys(user);
        driver.findElement(password).sendKeys(pass);
        driver.findElement(login).click();
    }
}
```

Q64. Disadvantages of PageFactory?

Interview-ready answer: Proxy-based lifecycle can be less explicit, especially with dynamic DOMs. By locators often give clearer control. Java / implementation example public final class WaitUtils { private final WebDriverWait wait; public WaitUtils(WebDriver driver) { wait = new WebDriverWait(driver, Duration.ofSeconds(10)); } public WebElement visible(By locator) { return wait.until(ExpectedConditions.visibilityOfElementLocated(locator)); } } Worldline Selenium Interview Guide | Java Page 14

Java / implementation example

```
public final class WaitUtils {
    private final WebDriverWait wait;

    public WaitUtils(WebDriver driver) {
        wait = new WebDriverWait(driver, Duration.ofSeconds(10));
    }

    public WebElement visible(By locator) {
        return wait.until(
            ExpectedConditions.visibilityOfElementLocated(locator));
    }
}
```

Q65. How do you handle stale elements with PageFactory?

Interview-ready answer: Avoid long-lived element references; re-locate and synchronise with the current DOM. Java / implementation example By status = By.id("status"); // Re-locate after the DOM updates; do not reuse an old WebElement. WebElement current = wait.until(ExpectedConditions.visibilityOfElementLocated(status)); System.out.println(current.getText());

Java / implementation example

```
By status = By.id("status");
// Re-locate after the DOM updates; do not reuse an old WebElement.
WebElement current = wait.until(
    ExpectedConditions.visibilityOfElementLocated(status));
System.out.println(current.getText());
```

Q66. Should Page Objects contain assertions?

Interview-ready answer: Keep most assertions in validation/test layers; page state helpers such as isLoggedIn can be useful. Java / implementation example // Page object exposes state; test owns the assertion. String message = loginPage.getErrorMessage(); Assert.assertEquals(message, "Invalid username or password");

Java / implementation example

```
String actual = driver.findElement(By.id("status")).getText();
Assert.assertEquals(actual, "Success", "Unexpected status displayed");
```

Q67. How do you avoid huge Page Objects?

Interview-ready answer: Create reusable component objects for headers, tables, menus and widgets. Java / implementation example public class HeaderComponent { private final WebDriver driver; private final By accountMenu = By.id("account-menu"); public HeaderComponent(WebDriver driver) { this.driver = driver; } public void openAccountMenu() { driver.findElement(accountMenu).click(); } } Worldline Selenium Interview Guide | Java Page 15

Java / implementation example

```
public class HeaderComponent {
    private final WebDriver driver;
    private final By accountMenu = By.id("account-menu");

    public HeaderComponent(WebDriver driver) {
        this.driver = driver;
    }

    public void openAccountMenu() {
        driver.findElement(accountMenu).click();
    }
}
```

9. Framework Architecture & Reusability

Q68. Explain scalable Selenium framework architecture.

Interview-ready answer: Separate tests, pages/components, driver lifecycle, config, data, utilities, listeners/reporting and CI concerns. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q69. Where should WebDriver be created?

Interview-ready answer: In a central driver factory/lifecycle layer; inject/pass it to page objects. Java / implementation example public final class DriverFactory { private DriverFactory() {} public static WebDriver create() { ChromeOptions options = new ChromeOptions(); options.addArguments("--start-maximized"); return new ChromeDriver(options); } }

Java / implementation example

```
public final class DriverFactory {
    private DriverFactory() {}

    public static WebDriver create() {
        ChromeOptions options = new ChromeOptions();
        options.addArguments("--start-maximized");
        return new ChromeDriver(options);
    }
}
```

Q70. How do you manage environments?

Interview-ready answer: Externalise URLs, browser, timeouts and other configuration; do not hardcode environment values. Java / implementation example String baseUrl = System.getProperty("baseUrl", System.getenv().getOrDefault("BASE_URL", "https://test.example.com")); driver.get(baseUrl);

Java / implementation example

```
String baseUrl = System.getProperty(
    "baseUrl", "https://test.example.com");
driver.get(baseUrl);
```

Q71. How do you manage secrets?

Interview-ready answer: Use Jenkins/cloud secret stores/environment variables; never hardcode credentials in source. Java / implementation example String username = System.getenv("TEST_USERNAME"); String password = System.getenv("TEST_PASSWORD"); if (username == null || password == null) { throw new IllegalStateException("Configure test credentials in CI secrets"); }

Java / implementation example

```
String username = System.getenv("TEST_USERNAME");
String password = System.getenv("TEST_PASSWORD");
if (username == null || password == null) {
    throw new IllegalStateException("Configure test credentials in CI secrets");
}
```

Q72. How do you design reusable utilities?

Interview-ready answer: One responsibility per utility; avoid a giant catch-all Utils class. Worldline Round 2 — Selenium Comprehensive Interview Preparation Page 8 Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q73. How do you integrate with Jenkins?

Interview-ready answer: Run through Maven/Gradle, pass environment parameters, publish reports and archive diagnostics. Java / implementation example // Jenkins pipeline step: sh "mvn -B clean test -DbaseUrl=$TEST_BASE_URL" // Publish Surefire XML and archive screenshots/logs in post { always { ... } }.

Java / implementation example

```
// Jenkins pipeline step:
sh "mvn -B clean test -DbaseUrl=$TEST_BASE_URL"
// Publish Surefire XML and archive screenshots/logs in post { always { ... } }.
```

Q74. How do you integrate reporting?

Interview-ready answer: Use Allure/Extent or equivalent listeners/hooks and attach screenshots/logs/environment details. Java / implementation example System.out.println("URL: " + driver.getCurrentUrl()); System.out.println("Title: " + driver.getTitle()); Files.writeString(Path.of("target/page.html"), driver.getPageSource()); Worldline Selenium Interview Guide | Java Page 16

Java / implementation example

```
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + driver.getTitle());
Files.writeString(Path.of("target/page.html"), driver.getPageSource());
```

Q75. How do you reduce framework flakiness?

Interview-ready answer: Stable locators, explicit waits, isolated data, deterministic setup, clean driver lifecycle and controlled parallelism. Java / implementation example WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement button = wait.until(ExpectedConditions.elementToBeClickable(By.id("submit"))); button.click();

Java / implementation example

```
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
WebElement button = wait.until(
    ExpectedConditions.elementToBeClickable(By.id("submit")));
button.click();
```

Q76. How do you review automation code?

Interview-ready answer: Review locators, waits, assertions, exceptions, independence, duplication, security and maintainability. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed()); Worldline Selenium Interview Guide | Java Page 17

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

10. Parallel Execution & Selenium Grid

Q77. How do you run Selenium tests in parallel safely?

Interview-ready answer: Independent WebDriver sessions, isolated data and thread-safe lifecycle. Java / implementation example private static final ThreadLocal<WebDriver> DRIVER = new ThreadLocal<>(); public static WebDriver getDriver() { return DRIVER.get(); } public static void quitDriver() { WebDriver driver = DRIVER.get(); try { if (driver != null) driver.quit(); } finally { DRIVER.remove(); } }

Java / implementation example

```
private static final ThreadLocal<WebDriver> DRIVER = new ThreadLocal<>();

public static WebDriver getDriver() {
    return DRIVER.get();
}

public static void quitDriver() {
    WebDriver driver = DRIVER.get();
    try {
        if (driver != null) driver.quit();
    } finally {
        DRIVER.remove();
    }
}
```

Q78. Why ThreadLocal?

Interview-ready answer: It gives each thread its own driver reference and prevents parallel tests from controlling the same session. Java / implementation example private static final ThreadLocal<WebDriver> DRIVER = new ThreadLocal<>(); public static WebDriver getDriver() { return DRIVER.get(); } public static void quitDriver() { WebDriver driver = DRIVER.get(); try { if (driver != null) driver.quit(); } finally { DRIVER.remove(); } }

Java / implementation example

```
private static final ThreadLocal<WebDriver> DRIVER = new ThreadLocal<>();

public static WebDriver getDriver() {
    return DRIVER.get();
}

public static void quitDriver() {
    WebDriver driver = DRIVER.get();
    try {
        if (driver != null) driver.quit();
    } finally {
        DRIVER.remove();
    }
}
```

Q79. How do you clean up ThreadLocal?

Interview-ready answer: Quit the driver and call ThreadLocal.remove() in teardown. Java / implementation example private static final ThreadLocal<WebDriver> DRIVER = new ThreadLocal<>(); public static WebDriver getDriver() { return DRIVER.get(); } public static void quitDriver() { WebDriver driver = DRIVER.get(); try { if (driver != null) driver.quit(); } finally { DRIVER.remove(); } }

Java / implementation example

```
private static final ThreadLocal<WebDriver> DRIVER = new ThreadLocal<>();

public static WebDriver getDriver() {
    return DRIVER.get();
}

public static void quitDriver() {
    WebDriver driver = DRIVER.get();
    try {
        if (driver != null) driver.quit();
    } finally {
        DRIVER.remove();
    }
}
```

Q80. What causes parallel failures?

Interview-ready answer: Shared drivers/data, static state, order dependencies, infrastructure limits and application race conditions. Java / implementation example String uniqueEmail = "qa+" + UUID.randomUUID() + "@example.test"; // Use unique test data to avoid collisions during parallel execution.

Java / implementation example

```
String uniqueEmail = "qa+" + UUID.randomUUID() + "@example.test";
// Use unique test data to avoid collisions during parallel execution.
```

Q81. How do you decide what can run in parallel?

Interview-ready answer: Run independent tests in parallel; redesign or serialise flows with shared state. Java / implementation example String uniqueEmail = "qa+" + UUID.randomUUID() + "@example.test"; // Use unique test data to avoid collisions during parallel execution.

Java / implementation example

```
String uniqueEmail = "qa+" + UUID.randomUUID() + "@example.test";
// Use unique test data to avoid collisions during parallel execution.
```

Q82. How do you size Grid parallelism?

Interview-ready answer: Measure CPU, memory, network, application capacity and queue time; increase concurrency based on evidence. Java / implementation example // Example TestNG configuration in testng.xml: // <suite name="UI suite" parallel="tests" thread-count="4"> // Keep each test's driver and test data independent. Worldline Selenium Interview Guide | Java Page 18

Java / implementation example

```
// Example TestNG configuration in testng.xml:
// <suite name="UI suite" parallel="tests" thread-count="4">
// Keep each test's driver and test data independent.
```

Q83. What is RemoteWebDriver?

Interview-ready answer: A WebDriver client that creates/controls a browser session on a remote Selenium server/Grid. Java / implementation example URL gridUrl = URI.create("http://localhost:4444").toURL(); WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions()); driver.get("https://example.com");

Java / implementation example

```
URL gridUrl = URI.create("http://localhost:4444").toURL();
WebDriver driver = new RemoteWebDriver(gridUrl, new ChromeOptions());
driver.get("https://example.com");
```

Q84. How do you troubleshoot Grid failures?

Interview-ready answer: Check nodes, capacity, browser/driver compatibility, network, Grid logs and infrastructure health. Java / implementation example System.out.println("URL: " + driver.getCurrentUrl()); System.out.println("Title: " + driver.getTitle()); Files.writeString(Path.of("target/page.html"), driver.getPageSource()); Worldline Selenium Interview Guide | Java Page 19

Java / implementation example

```
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + driver.getTitle());
Files.writeString(Path.of("target/page.html"), driver.getPageSource());
```

11. Flaky Tests & Debugging Strategy

Q85. What is a flaky test?

Interview-ready answer: A test whose result is inconsistent without an intentional application/test change. Java / implementation example try { runTestFlow(); } catch (RuntimeException e) { byte[] png = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES); Files.write(Path.of("target/failure.png"), png); Files.writeString(Path.of("target/page-source.html"), driver.getPageSource()); throw e; }

Java / implementation example

```
try {
    runTestFlow();
} catch (RuntimeException e) {
    byte[] png = ((TakesScreenshot) driver).getScreenshotAs(OutputType.BYTES);
    Files.write(Path.of("target/failure.png"), png);
    Files.writeString(Path.of("target/page-source.html"), driver.getPageSource());
    throw e;
}
```

Q86. Why does a test fail only in CI?

Interview-ready answer: Compare browser/configuration, timing, data, parallelism, network/environment and application behaviour. Java / implementation example ChromeOptions options = new ChromeOptions(); options.addArguments("--headless=new", "--window-size=1440,1000"); WebDriver driver = new ChromeDriver(options);

Java / implementation example

```
ChromeOptions options = new ChromeOptions();
options.addArguments("--headless=new", "--window-size=1440,1000");
WebDriver driver = new ChromeDriver(options);
```

Q87. What if a test passes locally but fails headless? Worldline Round 2 — Selenium Comprehensive Interview Preparation Page 9

Interview-ready answer: Compare headless config, viewport, rendering, timing, permissions and environment; do not blindly add sleeps. Java / implementation example ChromeOptions options = new ChromeOptions(); options.addArguments("--headless=new", "--window-size=1440,1000"); WebDriver driver = new ChromeDriver(options);

Java / implementation example

```
ChromeOptions options = new ChromeOptions();
options.addArguments("--headless=new", "--window-size=1440,1000");
WebDriver driver = new ChromeDriver(options);
```

Q88. What if it fails after 20 successful runs?

Interview-ready answer: Capture evidence and investigate data, environment, recent changes, resource leaks and timing. Java / implementation example String uniqueEmail = "qa+" + UUID.randomUUID() + "@example.test"; // Use unique test data to avoid collisions during parallel execution.

Java / implementation example

```
String uniqueEmail = "qa+" + UUID.randomUUID() + "@example.test";
// Use unique test data to avoid collisions during parallel execution.
```

Q89. Should flaky tests be retried?

Interview-ready answer: Only in a controlled, observable way. Indiscriminate retry hides defects. Java / implementation example // Use a runner-level retry policy with a small, bounded attempt count. // Record both the original failure and retry result; do not hide flakiness.

Java / implementation example

```
// Use a runner-level retry policy with a small, bounded attempt count.
// Record both the original failure and retry result; do not hide flakiness.
```

Q90. How do you measure automation quality?

Interview-ready answer: Stable pass rate, flake rate, execution time, meaningful coverage, escaped defects and diagnosis/maintenance effort. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed()); Worldline Selenium Interview Guide | Java Page 20

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

12. Senior SDET / Lead Scenarios

Q91. How would you improve an existing Selenium framework?

Interview-ready answer: Baseline flakiness, runtime, coverage and maintenance; then improve locators, waits, data, lifecycle, components, reporting and parallelism. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q92. How would you migrate Selenium to Playwright?

Interview-ready answer: Define measurable goals, pilot, migrate incrementally, compare runtime/flakiness and retire old code after confidence. Java / implementation example // Selenium Java:
wait.until(ExpectedConditions.elementToBeClickable(By.id("save"))).click(); // Playwright Java equivalent: //
page.locator("#save").click();

Java / implementation example

```
// Selenium Java:
wait.until(ExpectedConditions.elementToBeClickable(By.id("save"))).click();
// Playwright Java equivalent:
// page.locator("#save").click();
```

Q93. Developer says the test is wrong. What do you do?

Interview-ready answer: Stay evidence-driven: reproduce, compare requirements and actual behaviour, share diagnostics and resolve ambiguity with product. Java / implementation example String actual = driver.findElement(By.id("status")).getText();
Assert.assertEquals(actual, "Success", "Unexpected status displayed");

Java / implementation example

```
String actual = driver.findElement(By.id("status")).getText();
Assert.assertEquals(actual, "Success", "Unexpected status displayed");
```

Q94. Release has failing automation. What do you do?

Interview-ready answer: Classify failures by product/automation/environment/data and communicate business risk with evidence. Java / implementation example System.out.println("URL: " + driver.getCurrentUrl()); System.out.println("Title: " + driver.getTitle()); Files.writeString(Path.of("target/page.html"), driver.getPageSource());

Java / implementation example

```
System.out.println("URL: " + driver.getCurrentUrl());
System.out.println("Title: " + driver.getTitle());
Files.writeString(Path.of("target/page.html"), driver.getPageSource());
```

Q95. How do you decide what to automate?

Interview-ready answer: Prioritise stable, repeatable, high-value regression and critical business flows. Java / implementation example // Example risk-based test matrix: Object[][] cases = { {"valid user", "correct password", true}, {"valid user", "wrong password", false}, {"unknown user", "any password", false} };

Java / implementation example

```
Object[][] cases = {
    {"valid user", "correct password", true},
    {"valid user", "wrong password", false},
    {"unknown user", "any password", false}
};
```

Q96. How do you improve coverage?

Interview-ready answer: Use risk-based positive/negative/boundary/role/integration scenarios, not raw test-count targets. Java / implementation example // Example risk-based test matrix: Object[][] cases = { {"valid user", "correct password", true}, {"valid user", "wrong password", false}, {"unknown user", "any password", false} };

Java / implementation example

```
Object[][] cases = {
    {"valid user", "correct password", true},
    {"valid user", "wrong password", false},
    {"unknown user", "any password", false}
};
```

Q97. How do you mentor junior engineers?

Interview-ready answer: Teach reasoning and framework principles through pairing, code reviews and progressively increasing ownership. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed()); Worldline Selenium Interview Guide | Java Page 21

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q98. How do you handle technical disagreement?

Interview-ready answer: Define criteria, compare evidence, run a small proof of concept if needed and document the decision. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Q99. How would you reduce a four-hour regression suite?

Interview-ready answer: Profile first; remove duplication, improve data/setup, parallelise independent tests and optimise expensive operations. Java / implementation example // Example TestNG configuration in testng.xml: // <suite name="UI suite" parallel="tests" thread-count="4"> // Keep each test's driver and test data independent.

Java / implementation example

```
// Example TestNG configuration in testng.xml:
// <suite name="UI suite" parallel="tests" thread-count="4">
// Keep each test's driver and test data independent.
```

Q100. What makes you a Senior SDET rather than only an automation coder?

Interview-ready answer: Quality engineering: risk-based testing, maintainable automation, debugging, CI feedback, mentoring and engineering decisions based on evidence. Java / implementation example // Illustrative framework pattern: keep actions explicit, wait for state, // assert expected outcomes, and retain useful diagnostics on failure. wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready"))); Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed()); Worldline Selenium Interview Guide | Java Page 22

Java / implementation example

```
// Illustrative framework pattern: wait for state, act, then assert.
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ready")));
Assert.assertTrue(driver.findElement(By.id("ready")).isDisplayed());
```

Final Revision Checklist

- Prefer stable, unique locators and meaningful test attributes.
- Use explicit, condition-based waits instead of fixed sleeps.
- Re-locate elements after DOM updates; investigate stale references.
- Investigate overlays before using JavaScript interactions.
- Switch frames and windows deliberately and restore context.
- Keep Page Objects focused; keep most assertions in the test layer.
- Use one WebDriver per parallel test and clean up ThreadLocal values.
- Capture screenshots, page source, URL, and logs on failure.
- Keep credentials in environment variables or a secret manager.
- Measure flakiness, execution time, risk coverage, and maintenance effort.
- Use bounded, visible retries; never hide a failure.
- For senior answers, explain trade-offs, failure modes, CI behaviour, and maintainability.

Common Java imports

```
import java.io.IOException;
import java.net.URI;
import java.net.URL;
import java.nio.file.Files;
import java.nio.file.Path;
import java.time.Duration;
import java.util.UUID;

import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.FindBy;
import org.openqa.selenium.support.ui.*;
import org.testng.Assert;
```

Imports vary by snippet. Verify browser-specific CDP/BiDi APIs against the Selenium and browser versions used by your project.